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PALEOMAGNETISM OF THE LATE ORDOVICIAN THOUARS MASSIF, VENDEE PROVINCE, FRANCE

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Abstract. A paleomagnetic study of the Late Ordovician Thouars Massif reveals very steep components of magnetization, directed upward and downward, and with thermally discrete blocking temperatures of 550° to 580°C. This characteristic magnetization yields a mean direction with declination/inclination = 340°/-83°, with k = 27 and α_{95}=7°, based on 15 sites. The south paleopole is located at 34°N, 5°E. A partial magnetic overprint of Hercynian age is observed in sites close to the South Armorican Shear Zone. The 15 sites are spread over some 20 by 30 km, and the internal consistency of the site-mean directions indicates that differential tilting of parts of the massif has not occurred. Since it appears to be unlikely that the entire massif was tilted, we infer that the characteristic direction is primary and that the paleopole adequately represents the local magnetic field in the Late Ordovician. The paleopole for the Thouars Massif supports previously obtained results from other Ordovician rocks in Hercynian Europe, but it is the first result to pass modern reliability criteria. The high Ordovician paleolatitudes of the Armorican Massif, other areas in Hercynian Europe, and northwestern Africa, indicate that Gondwana and Armorica were adjacent until Late Ordovician time. Low paleolatitudes for Late Devonian time suggest that Armorica underwent rapid northward motion, at a rate of about 10 cm/yr, during the Silurian and earliest Devonian. Armorica is thought to have collided with North America and the Baltic shield/Russian Platform before or during the Early Devonian, forming the Old Red Continent. The collisions produced the Caledonian and Acadian orogenic belts.

Introduction

Plate tectonic models for Paleozoic time have been proposed in the last few years on the basis of paleomagnetic data as well as geological similarities and dissimilarities between various blocks. One of these models involves an independent plate called Armorica, consisting of Hercynian Europe and possibly the Avalonian basement blocks in eastern North America. Armorica traveled, after the Cambrian but before the Late Devonian, away from Gondwana and toward the northern cratons of North America and the combined Baltic Shield-Russian Platform [Van der Voo, 1979, 1982, 1983; Perroud and Bonhommet, 1981]. A more precise timing of this northward drift of Armorica is of interest, because it may resolve the ambiguities in choosing between the Taconic and Acadian orogenies as the principal collisional event between Armorica and North America.

Although a few paleomagnetic data have become available for Ordovician rocks from Armorica [Duff, 1979; Perroud and Bonhommet, 1981; Perroud et al., 1983], uncertainties remain about the age and direction of magnetization, whereas for Silurian time, no data of sufficient reliability are available. In order to obtain a better resolution of the Ordovician geomagnetic field direction in the Armorican Massif, sampling and paleomagnetic measurements were initiated in 1981 with a collection from the Late Ordovician Thouars Massif, Vendée province, France.

Geological Setting and Sampling

The southern part of the Armorican Massif has been the site of a Siluro-Devonian orogenic phase, called the Ligerian orogeny [Cogné, 1977], and a paired metamorphic belt has been documented for this area [Lefort, 1979; Peucat et al., 1978]. The high-pressure, low-temperature metamorphism of the Ile de Groix gave rise to a blueschist facies with one of the better known pre-Mesozoic occurrences of glaucophane in the world [Carpenter and Civetta, 1976]. Immediately to the north of this paired metamorphic belt, the South Armorican Shear Zone (SASZ) is marked by ultra-mylonites and leucogranitic intrusions of Hercynian age; the SASZ has been interpreted as a dextral megashear of Carboniferous age [Jegouzo, 1980; Berthé et al., 1979]. Located just to the north of the "Cholet" branch of the SASZ, a hypovolcanic intrusive complex, the Thouars Massif, has been dated by Bernard-Griffiths and Le Métour [1979] with a Rb-Sr isochron as 444 ± 9 Ma, i.e., Late Ordovician in age.

The SASZ appears to have affected the southern part of the Thouars Massif, as seen in folded tectonic foliation and microveins, whereas the more northerly parts of the Massif show no evidence of deformation. The Thouars complex intruded into fossiliferous Cambro-Ordovician volcano-sedimentary sequences, and is overlain by rhyolites of the "Formation des Volcanites de la Plaine," which have been interpreted as the effusive series associated with the plutonic intrusions. The Massif itself consists of pink microgranites, with minor gabbroic or granodioritic phases of the same age, as shown by contact relationships between the phases [Bernard-Griffiths and Le Métour, 1979]. The Rb-Sr dating was done on microgranite of the "La Gouraudière" quarry, with a second set of samples from rhyolite in a breccia giving very similar ages; the two data sets have been

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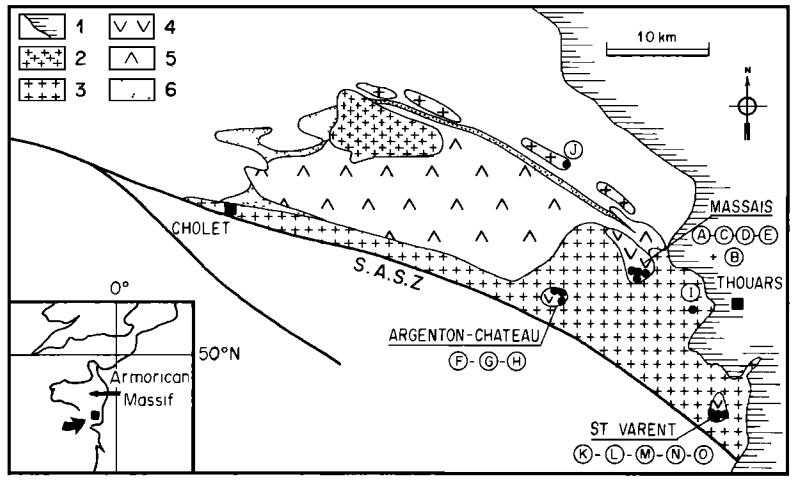


Fig. 1. Schematic geological map of the Thouars complex with sampling sites for this study indicated by letters. Inset shows the location of the complex within the Armorican Massif of France. SASZ is the Cholet branch of the South Armorican Shear Zone. Legend: 1, younger (Mesozoic) cover of the edge of the Paris and Aquitaine basins; 2, Late Hercynian granite; 3, microgranitic phases of the Thouars igneous complex; 4, gabbroic and granodioritic phases of the Thouars complex; 5, rhyolitic phases of the complex; and 6, Cambro-Ordovician sediments.

combined to yield one isochron with excellent precision.

Paleomagnetic sampling was concentrated in the more mafic phases of the Thouars complex, with 14 sites of gabbro, granodiorite, and dolerite. At one site (B), microgranite was sampled. Our sites include the quarry of La Gouraudière (site I) where the material for dating was obtained.

In intrusive complexes, such as the Thouars Massif, there is in general no possibility to evaluate structural tilting. However, our

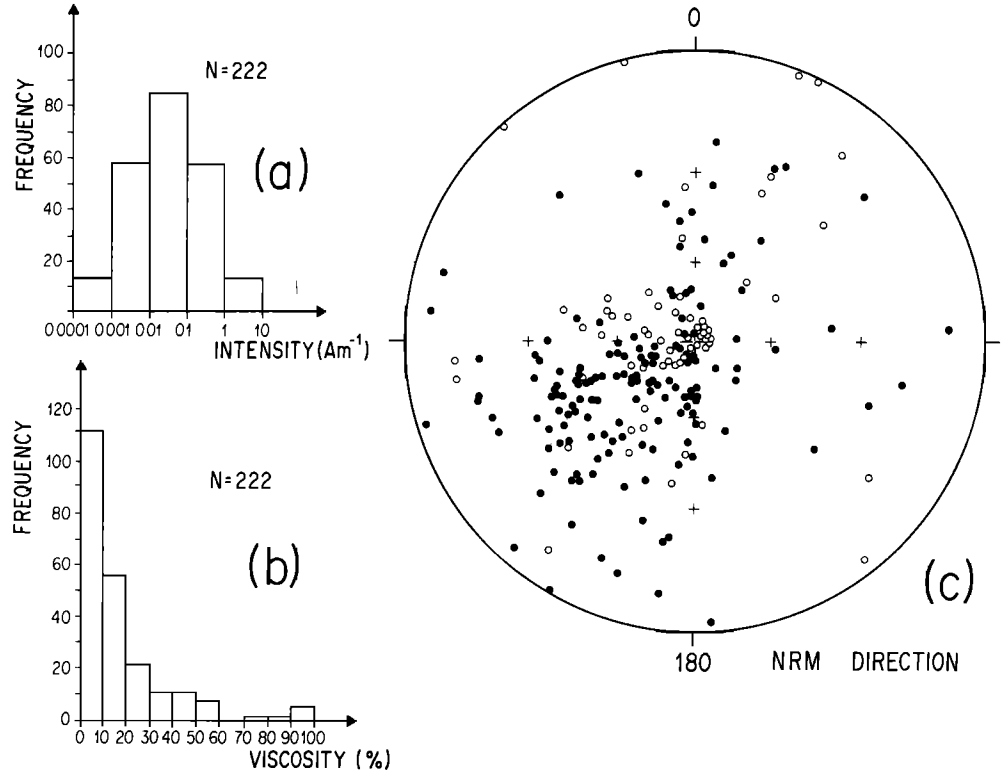


Fig. 2. (a) Frequency of NRM intensities observed in the Thouars samples, (b) frequency of viscosity coefficients of our samples [Thellier and Thellier, 1959], and (c) directions of NRM (before demagnetization) in stereographic projection, Open (closed) symbols denote upper (lower) hemisphere projections.

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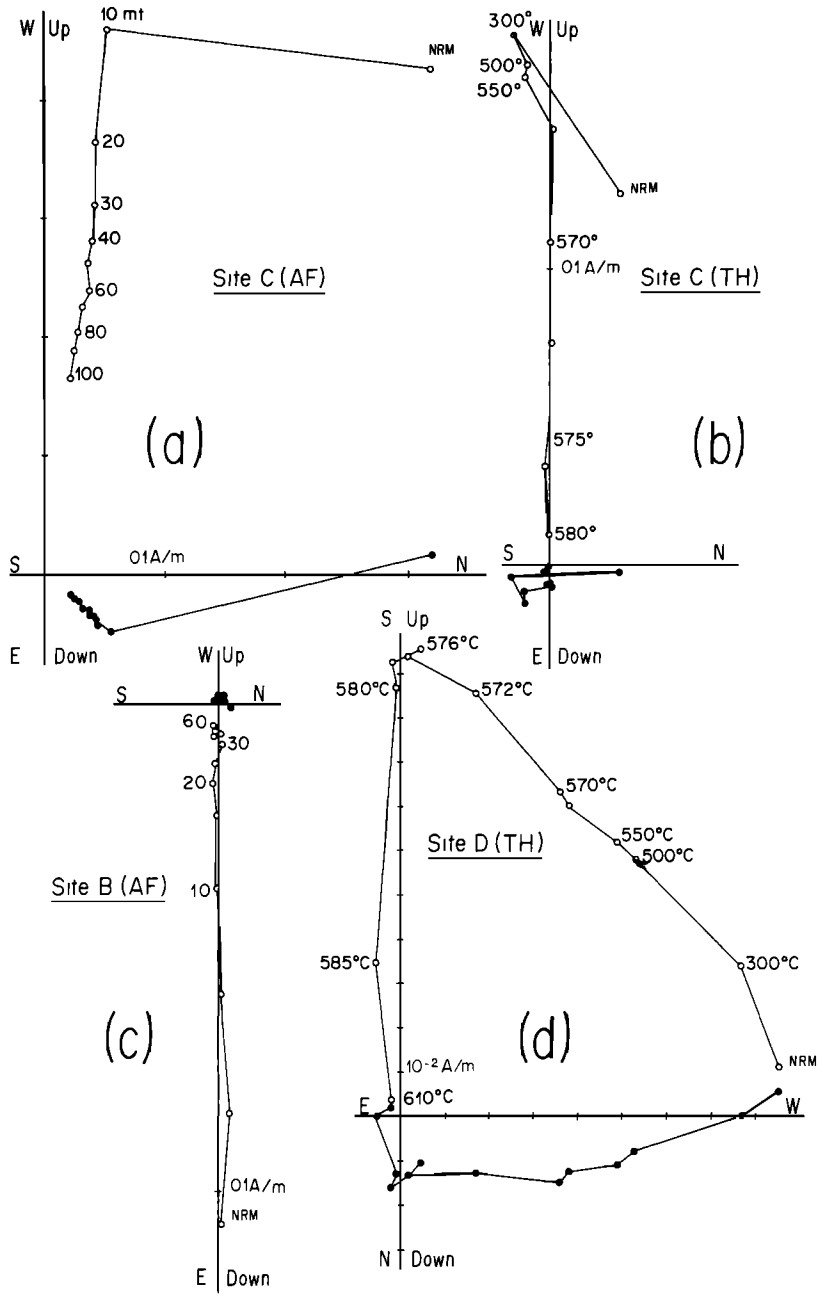


Fig. 3. Alternating field (AF) and thermal (TH) demagnetizations of samples from the Massais area (sites A-E), plotted in orthogonal projections [Zijderveld, 1967]. Plotted points represent the endpoints of the magnetization vector during progressive demagnetization, open symbols represent projections onto the vertical plane, and closed symbols represent projections onto the horizontal plane.

sampling sites are distributed over tens of kilometers in diameter (Figure 1), with two sites in quarries (site I: 47°N, 359.6°E and site J: 47.1°N, 359.4°E) and the others in three clusters of more mafic phases. These gabbroic-granodioritic areas are named Massais (47°N, 359.5°E), Argenton-Chateau (47°N, 359.4°E), and Saint-Varent (46.9°N, 359.6°E). The distribution of the sites should allow us to place some constraints, albeit weak, on differential structural tilts within the Massif, as well as on a tilt of the complex as a whole.

Sampling was carried out with a portable field drill, and the cores were oriented with magnetic and solar compasses. A typical site had samples distributed over a 15 by 5 m surface area, with an average of nine samples per site. Individual sites, even those of the same gabbroic-granodioritic area, are separated by at least 300 m.

Laboratory Procedures and Results

The field-drilled cores were cut in the laboratory into specimens of 2.2 cm height. Natural remanent magnetizations (NRM's) have been

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TABLE 1. Site-Mean Characteristic Directions From the Thouars Massif

Site	Site Coordinates	N_s	n/n_o	Cleaning Range AF, mT	Cleaning Range Th, °C	Magnetic direction Declination	Magnetic direction Inclination	k	α_{95}	Polarity	VGP
A	47.0N,359.5E	7	11/11	30-60	450-575	238	-86	82	5	N	50.7S/190.2E
B	47.0N,359.5E	4	5/6	10-30	400-450	309	+75	22	17	R	57.9N/315.8E
C	47.0N,359.5E	11	17/17	30-80	570-580	38	-89	187	3	N	45.4S/177.7E
D	47.0N,359.5E	9	17/22	70-100	570-610	23	-81	103	4	N	30.5S/171.6E
E	47.0N,359.5E	10	8/16	20-80	400-550	211	-82	270	4	N	59.6S/195.5E
F	47.0N,359.4E	13	9/13	20-90	400-580	173	+63	20	14	R	1.7N/ 4.4E
G	47.0N,359.4E	11	15/20	20-80	500-580	201	+67	27	8	R	8.4N/345.8E
H	47.0N,359.4E	9	17/17	10-80	500-580	225	+87	25	7	R	42.6N/353.7E
I[1]	47.0N,359.6E	10	8/13	20-50	400-570	317	-82[1]	73	7	N	34.7S/192.6E
I[1]	47.0N,359.6E	10	4/13	30-100		120	+75[1]			R	28.9N/ 27.5E
J	47.1N,359.4E	10	10/17	30-100	560-580	229	-70	90	5	N	58.8S/238.3E
K	46.9N,359.6E	12	11/19	10-50	540-580	18	-69	39	8	N	10.6S/168.6E
K	46.9N,359.6E	12	5/19	10-50	200-540	182	+72	39	13	R	13.9N/358.5E
L	46.9N,359.6E	7	11/11	20-100	500-550	76	+78	78	5	R	47.4N/ 33.7E
M	46.9N,359.6E	8	8/11	20-80	400-580	233	+73	29	11	R	24.1N/332.4E
N_2	46.9N,359.6E	7	4/8	10-40	540-570	339	-62	19	22	N	2.0S/194.7E
O^2	46.9N,359.6E	7	7/8	10-70	540-570	274	-68	11	19	N+R[2]	32.5S/227.7E
Mean All Sites		17				160	+83	27	7	N+R	
Mean R Sites		8				186	+81	26	11	R	
Mean N Sites		9				310	-83	28	10	N	
Mean All VGP'S		17						9	13	N+R	34.4N/ 5.1E
Mean R VGP'S		8						8	21	R	29.9N/357.4E
Mean N VGP's		9						9	18	N	38.2S/192.6E

N_s is the number of samples per site or the number of sites used to compute the mean; n/n_o is the ratio of specimen directions used to compute the mean to the total number of specimens analyzed; AF is alternating fields, Th is thermal treatment; declination and inclination are in degrees; k and α_{95} are the statistical parameters associated with the mean [Fisher, 1953] and the polarity interpretation is N for normal and R for reversed; VGP is virtual geomagnetic pole position computed from the magnetic direction.

Specimen directions have been rejected from site-mean computations, when demagnetization trajectories did not permit isolation of the characteristic magnetization through vector subtraction spanning at least three or more colinear points.

[1]This mean direction was determined from the intersection of great circles (Figure 7c); Q, the quality factor of the great-circle fit [Halls, 1976], is $1.5 \cdot 10^{-4}$.

[2]At this site, polarities have been combined because the number of R entries was insufficient for statistical treatment.

measured with a cryogenic ScT magnetometer or with a Schonstedt spinner magnetometer. Before demagnetization, a short-term viscosity test [Thellier and Thellier, 1959] was carried out. Alternating field and thermal demagnetizations were performed with Schonstedt equipment, with companion specimens being subjected to different treatments (alternating field, AF, or thermal, th) in order to compare coercivity and blocking-temperature spectra of the components of magnetization. Orthogonal projections of the demagnetization results were used to determine the directions of components of magnetization [Zijderveld, 1967]; in cases where overlapping coercivity or blocking-temperature spectra gave rise to a suspicion that complete separation of components was not achieved, we used great-circle analysis [Halls, 1976] to establish the characteristic direction. Isothermal remanent magnetization (IRM) acquisition experiments have been performed with a cryogenic magnet, capable of reaching 5 T, and thermomagnetic analyses were done at the University of Rennes with a Brucker Curie balance. For a representative set of samples, polished thin-sections were examined in reflected light. NRM intensities (Figure 2a) are relatively low for such mafic rocks with a lognormal distribution and a modal peak intensity of 5×10^{-2} Am^{-1}. The short-period (8-day) viscosity coefficients are also low, with values less than 20% for three-quarters of the sample population (Figure 2b). Where high values of the viscosity coefficient were observed, low-coercivity, often-random components of magnetization were generally common. The NRM directions (Figure 2c) cluster in the southwestern lower-hemisphere quadrant or around a vertically upward direction.

As mentioned earlier, our samples have been obtained in three localized areas of gabbroic-granodioritic phases, as well as from two quarries. Below, the demagnetization results

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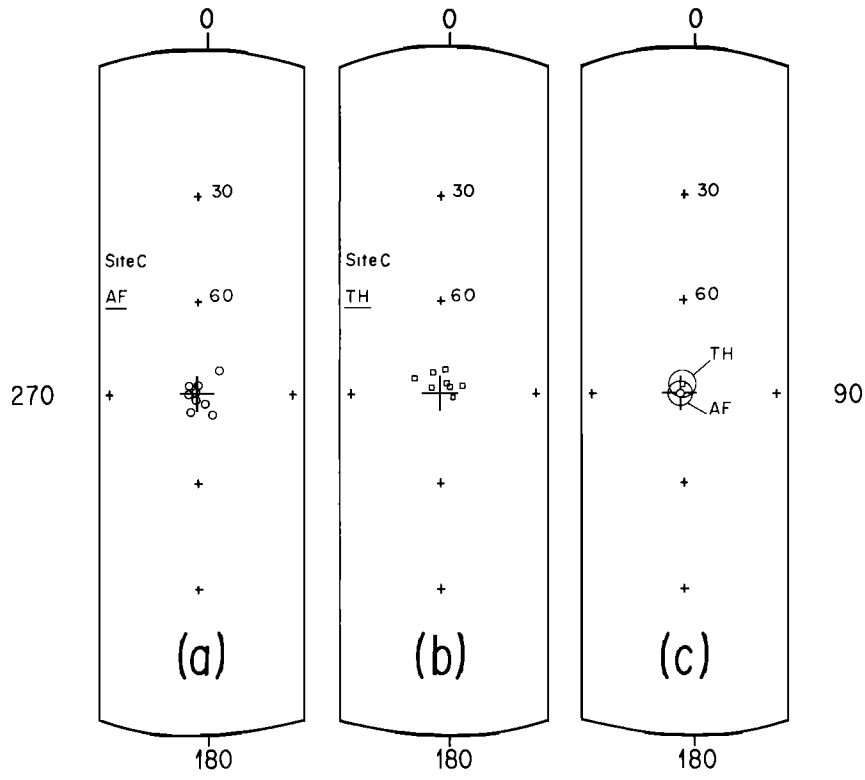


Fig. 4. Stereographic projections for the characteristic directions of magnetization of site C, obtained from (a) alternating field demagnetization or (b) thermal demagnetization. (c) The mean directions of these two groups and their cones of 95% confidence. Same conventions as in Figure 2c.

will be described for each of these four groups: Massais, Argenton-Chateau, the quarries, and Saint-Varent.

Massais (sites A-E)

In this area, which in contrast to two of the other groups of sites is not close to the SASZ, essentially univectorial decay of the magnetization was observed after removal of an occasional low-coercivity or low-blocking temperature component (Figures 3a-3d). AF demagnetization up to 100 mT removes well over half of the nearly vertically upward or downward component (Figures 3a and 3c). Thermal demagnetization experiments (Figure 3b) indicate blocking temperatures of this characteristic magnetization between 550° and 580°C. A few samples from site D show a two-component behavior, but the same characteristic component is removed above 580°C (Figure 3d).

Table 1 lists the mean directions, demagnetizing range, and statistical parameters associated with the means of the sites. For the Massais sites, few specimens had to be rejected (see n/n_o in Table 1), although site E formed an exception: here, observable weathering at the outcrop correlated with poor stability during demagnetization of half the specimens.

A few other observations for the Massais results are noteworthy. First, AF and thermal demagnetization for each site give nearly identical directions of magnetization, despite occasional soft overprinted components (Figure 4). Second, the characteristic magnetization decreases abruptly in intensity between 560° and 580°C (Figures 3b and 5b) for most samples, but the magnetization is not always completely removed until well above 600°C. If one computes, for instance, the direction for the samples of site D, the same result is obtained for the residual components above 580°C as for the components removed between 570° and 580°C (Figure 5c). This indicates to us that (1) hematite is occasionally the carrier of magnetization, (2) most likely magnetite is a carrier also, and as we will see later, this has been confirmed by optical and rock magnetic work, and (3) whatever the carrier of magnetization, the direction of the characteristic remanence is the same in these sites.

Argenton-Chateau (sites F-H)

NRM directions of these three sites are all in the lower-hemisphere southwest quadrant, and most specimens show upon demagnetization steeply downward characteristic magnetizations (see Table 1). However, shallower, often southwesterly directions were also removed during AF and thermal demagnetization, and these are compiled in Table 2. Site H is typical for this multicomponent behavior, which is illustrated in Figure 6. Above 560°C and between 20 and 100 mT, the characteristic (steeper) direction is removed (Figures 6b and 6c). A low-blocking temperature component is removed by 250°C (Figure 6c) and is interpreted as a present-day field direction acquired in the field or the

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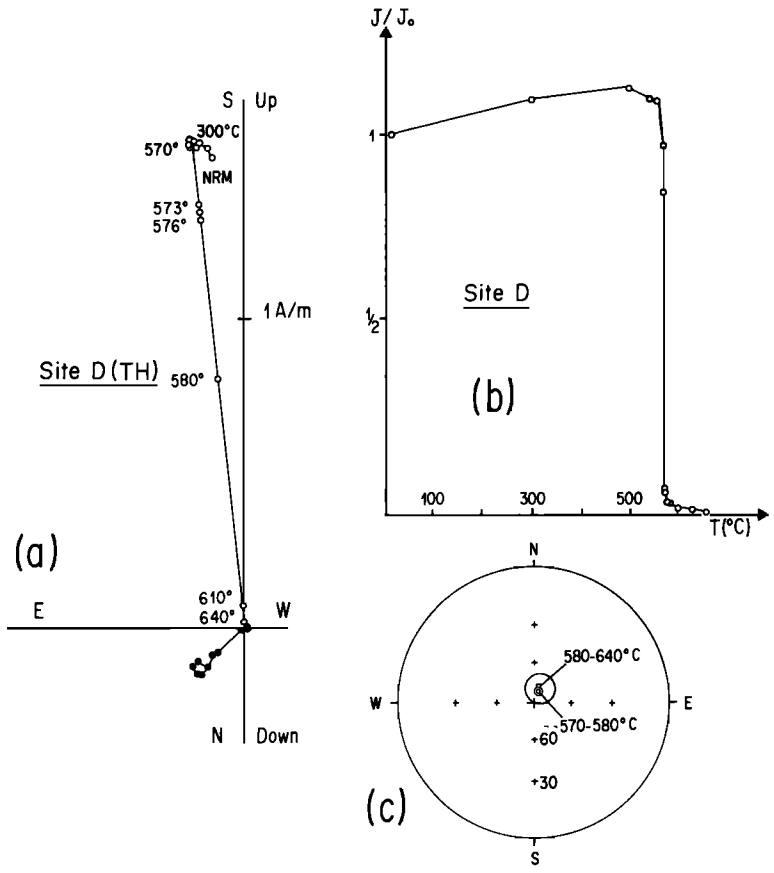


Fig. 5. Demagnetization results revealing the presence of hematite as well as magnetite in samples from site D. (a) Thermal demagnetization (as in Figure 3) showing blocking temperatures below as well as above 580°C. (b) Normalized intensity versus demagnetization temperature for another sample, showing pronounced decay in an interval typical of magnetite, as well as a very small remanence fraction with blocking temperatures characteristic of hematite. (c) Directions of the components of magnetization during thermal demagnetization averaged for this site for temperatures below 580° as well as above 580°C, in stereographic projection. Same conventions as in Figures 2 and 3.

TABLE 2. Hercynian Overprint: Site-mean directions

Site	n	Cleaning Range AF, mT	Cleaning Range Th, C	Declination	Inclination	k	α_{95}
C	5	20-50	300-570	183	+38	29	14
F	7	>50	0-580	231	+26	89	6
G	11	>50	300-500	225	+18	72	7
H	15	>60	300-500	230	+20	34	7
J	7		300-560	207	+ 9	13	18
L	6	0-20	200-500	232	+ 6	31	19
M*	6	20-90	0-570	260	+25	13	19
Mean	6			219	+20	15	18

Symbols and abbreviations are explained in Table 1. The south-paleopole is located at 23°S, 316°E, whereas the mean Carboniferous reference pole for the Armorican Massif is located at 30°S, 326°E [Perroud et al., 1985]. These two poles are statistically indistinguishable.

*This site has been excluded from the computation of the mean direction.

laboratory. However, between 250° and 560°C, or above 100 mT in AF demagnetization, a southwesterly and shallow component is isolated (Figures 6b and 6c), with a direction typical for the Late Paleozoic in Hercynian Europe (Table 2).

The Quarries (sites I-J)

Gabbroic to granodioritic phases were sampled at sites I and J (Figure 1). The magnetic behavior is very similar to that of the Massais area, and nearly vertical (up or down) characteristic directions were obtained (Figure 7a, Table 1). Site J, in addition, shows evidence of the shallow southwesterly overprint between 300°C and 560°C, before the characteristic and steeply up direction is removed between 560°C and 580°C. In site I, a few samples showed a complex multicomponent magnetization. A Zijderveld diagram and great-circle plot (Figures 7b and 7c) reveal the successive removal of (1) a soft downward component, (2) an intermediate upward component between 20 and 50 mT, and (3) a second downward component above 60 mT. We interpet this phenomenon as caused by overlapping coercivity

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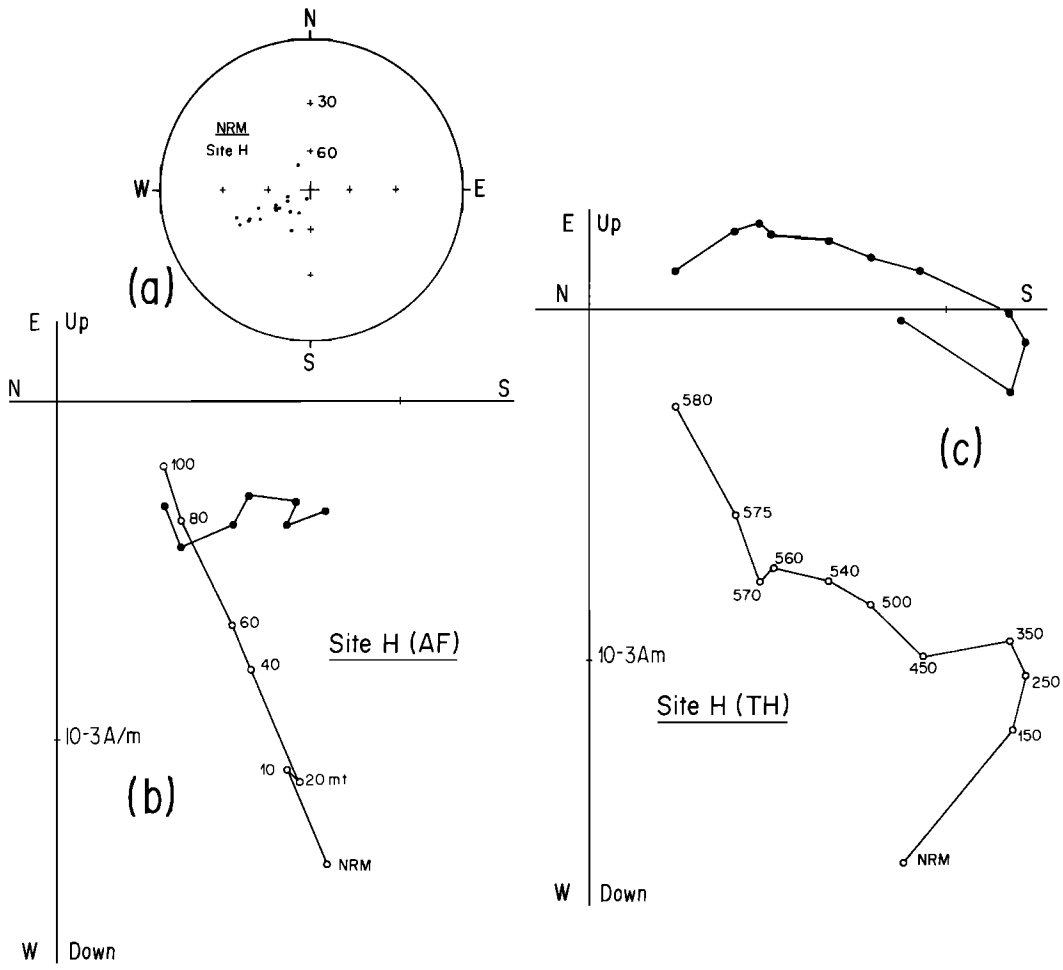


Fig. 6. (a) NRM directions, (b) alternating field demagnetization diagram, and (c) thermal demagnetization diagram of samples from site H (Argenton Chateau area). Same conventions as in Figures 2 and 3.

spectra, above 20 mT, of juxtaposed normal and reversed characteristic magnetizations; we will see that such behavior is much better documented in the Saint-Varent sites (K-O). Great-circle analysis of the trends shown in Figure 7c yields convergence at declination/inclination = 120°/+75° (see Table 1) for four specimens, in addition to the vector-subtraction results for an upward direction with declination/inclination = 317°/-82°.

Saint-Varent (sites K-O)

In the sites of this area, the viscosity coefficients (Figure 2b) were among the highest and, consequently, it came as no surprise to find more multicomponent behavior than in the previously discussed groups.

Sites K and L may serve as an example (Figures 8 and 9). In site K, low-coercivity (<30 mT) and low-blocking temperature (<500°C) components are northerly and down; these are interpreted as present-day field or laboratory-acquired viscous overprints. After removal of these components, however, eleven specimens showed a steep upward component (Figure 8c) and five other specimens yielded a steep and southerly downward component (Figure 8d) with near-antipodal directions (Table 1). Site L shows antipodal directions present in the same sample (Figure 9a), although in general only the steeply downward component (Figure 9b) is well isolated. The other components (Figure 9c) reveal a trend for site L toward a westerly and steeply upward direction, but clearly these components have not been determined accurately by vector-subtraction above 540°C and cannot be used for a paleopole determination.

Carriers of Magnetization

With occasional exceptions, such as noted earlier for blocking temperatures in the hematite range, the demagnetization results have all characteristics of magnetite. Especially, the sharp drop in intensity between 560° and 580°C (Figure 5b) seems to indicate relatively pure, fine-grained magnetite such as characteristically found as inclusions in the plagioclases of gabbroic samples. Median-coercive fields of about 80 mT support this contention, and imply a single-domain state [Dunlop et al., 1973].

To document further the nature of the remanence carriers, IRM acquisitions, thermomagnetic runs and optical petrography have been used, and all confirm the presence of magnetite,

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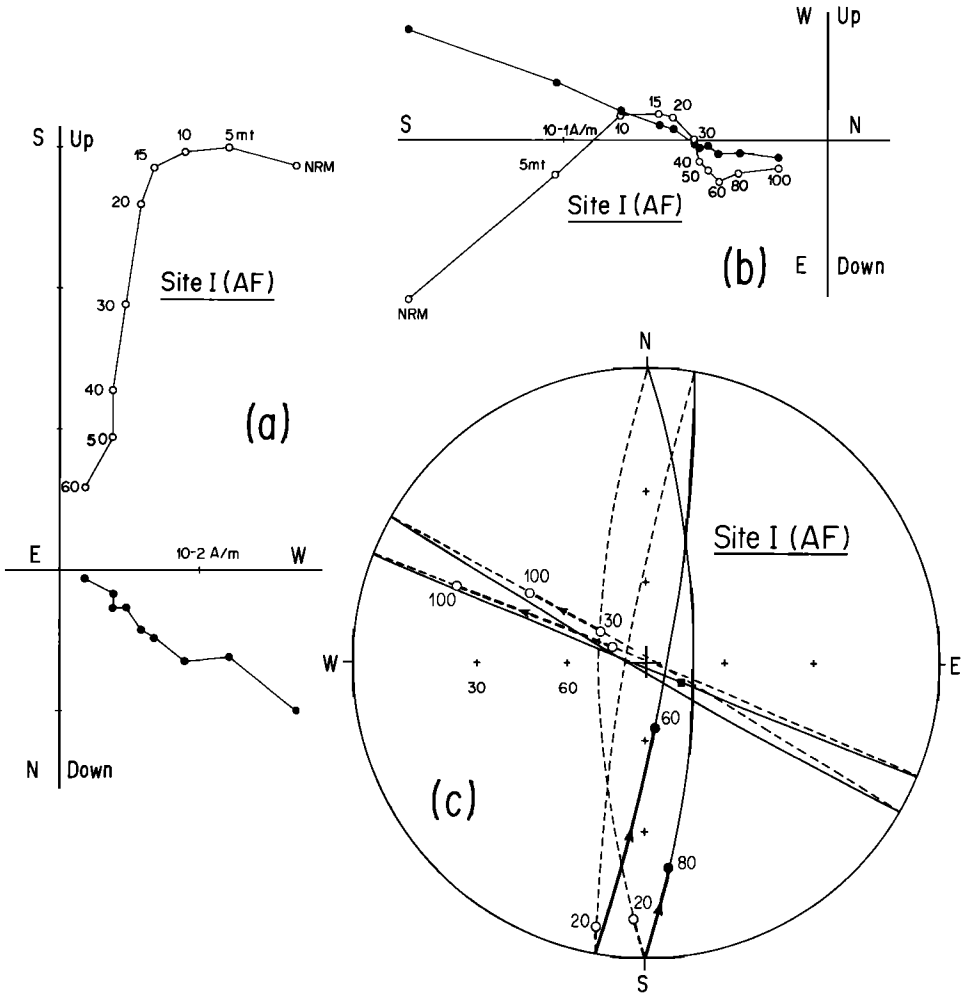


Fig. 7. Results from site I, the quarry of La Gouraudière. (a) Zijderveld diagram of an alternating field demagnetization (as in Figure 3), showing the characteristic and steeply upward direction; (b) alternating field demagnetization of another sample, showing superimposed components of magnetization with downward, upward, and again downward directions; (c) stereographic projection of the great-circles obtained from AF demagnetization of four samples from site I that show multivectorial behavior. Numbers refer to peak alternating fields (in mT); continuous (dashed) lines represent projections onto the lower (upper) hemisphere, with bold segments of these lines giving the observed trajectories. The square is the best-fit direction of the great-circle convergence.

as well as occasional hematite. Figure 10a displays an IRM acquisition curve for site F, characteristic of hematite, but with a fairly low-coercivity (<0.3T) backfield value. Figure 10b is more typical, with saturation reached below 0.2 T and suggests magnetite as the carrier of the remanence. A Lowrie-Fuller test [Lowrie and Fuller, 1971] on a gabbro sample from Massais indicates single-domain state. Figure 11 shows thermomagnetic analyses indicative of magnetite (Figures 11a-11c) or, in addition, small (Figure 11b) to sizeable but possibly ambiguous hematite contributions (Figure 11d). We recall that thermal demagnetization of samples from site D (Figure 3d) and IRM acquisition of site F (Figure 10a) also showed the presence of hematite. In general, the thermomagnetic curves of the gabbroic samples are reversible, but the microgranite of site B (Figure 11c) reveals some mineralogical changes.

Transmitted-light examination of thin-sections generally does not reveal traces of deformation or metamorphism, and the opaques appear primary. The latter is true even for samples from Argenton-Chateau, close to the SASZ, although some recrystallization and deformation can be seen there. Reflected-light microscopy shows that large (~100 μm) iron-oxide grains are common and that these have been exsolved into magnetite and ilmenite lamellae. This exsolution may have produced small grains, that fit the characteristics of the observed magnetizations. However, even if the large grains thus observed do not carry all or even part of the high-coercivity remanence, they nevertheless show that primary iron-oxide phases have survived in these samples. In the Argenton-Chateau samples (sites F-H), hematite is seen as alteration products of magnetites, in agreement with rock magnetic observations. Moreover, demagnetization revealed a high-coercivity compo-

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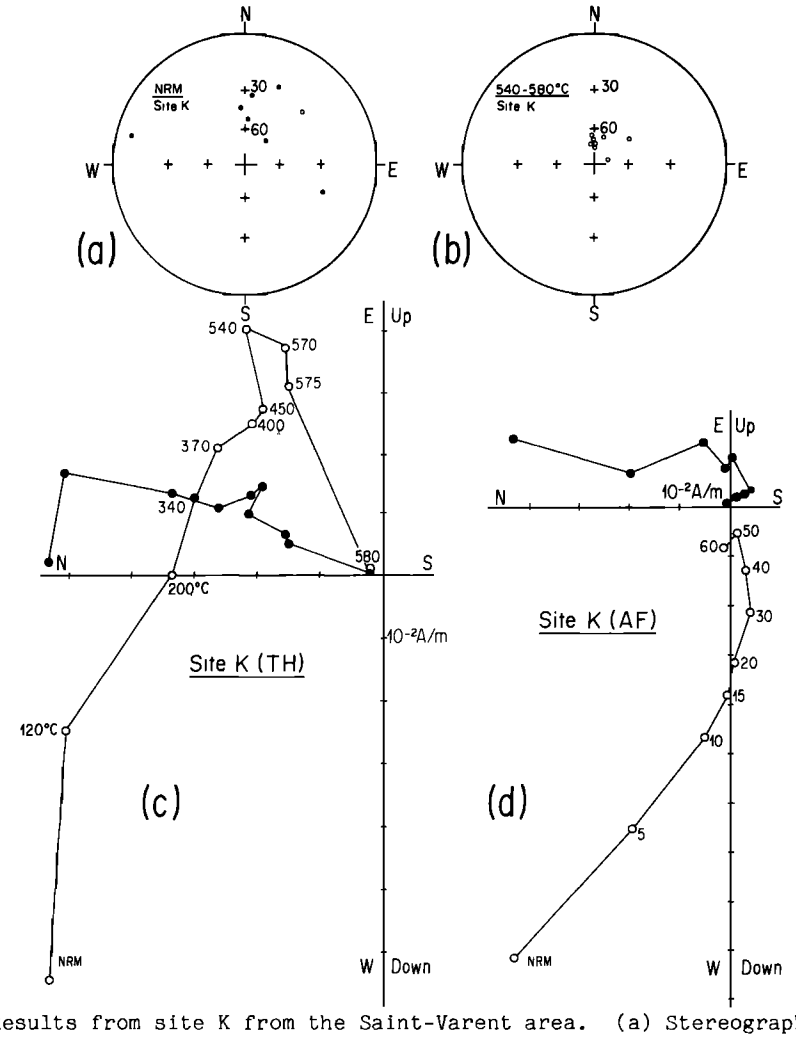


Fig. 8. Results from site K from the Saint-Varent area. (a) Stereographic plot with the NRM directions of thermally treated samples, (b) characteristic directions obtained during thermal demagnetization between 540° and 580°C, (c) thermal demagnetization diagram, and (d) alternating field demagnetization diagram (as in Figure 3). For site K, 11 specimens showed upward directions, as in Figure 8c, and five other specimens from different samples showed downward directions (as in Figure 8d), listed separately for each polarity in Table 1.

nent of magnetization which we interpret as a Late Paleozoic overprint. In view of this and of the deformation and recrystallization of samples in this area close to the SASZ, we infer a relationship with the Carboniferous movements along the fault.

We conclude that magnetite, fine grained and relatively free of titanium, is the predominant magnetic carrier. However, hematite is present in small amounts in site D and significantly in samples from the sites of Argenton-Chateau near the SASZ.

Discussion

The principal finding of this study consists of a steep, antipodal, and thermally discrete characteristic magnetization, carried generally by magnetite but also by hematite in site D. The mean direction with declination/inclination = 340°/-83° (Table 1) is unlike any direction observed in post-Ordovician rocks from the Armorican Massif, or from anywhere else in Europe for that matter. Thus we have reason to believe that we have obtained a primary magnetization, i.e., a magnetic direction of Late Ordovician age. Two observations further strengthen this contention.

The first observation concerns the antipodal nature of the characteristic remanence (Figure 12), which indicates that demagnetization has been effective in removing secondary overprints and has isolated a reliable record of the paleofield.

The second observation involves the internal consistency of the characteristic remanence between the four sampling areas or groups (Figure 13). This appears to rule out any differential structural tilts greater than 20° between parts of the intrusive complex. A tilt of the complex as a whole is more difficult to rule out, but the size of the Thouars Massif would appear to preclude any tilt of significance, say, much greater than 20°. This structural uncertainty which exists, of course, for all nonstratified bodies, implies that even when it is taken into account, an inclination greater than 60° is obtained. In view of other steep

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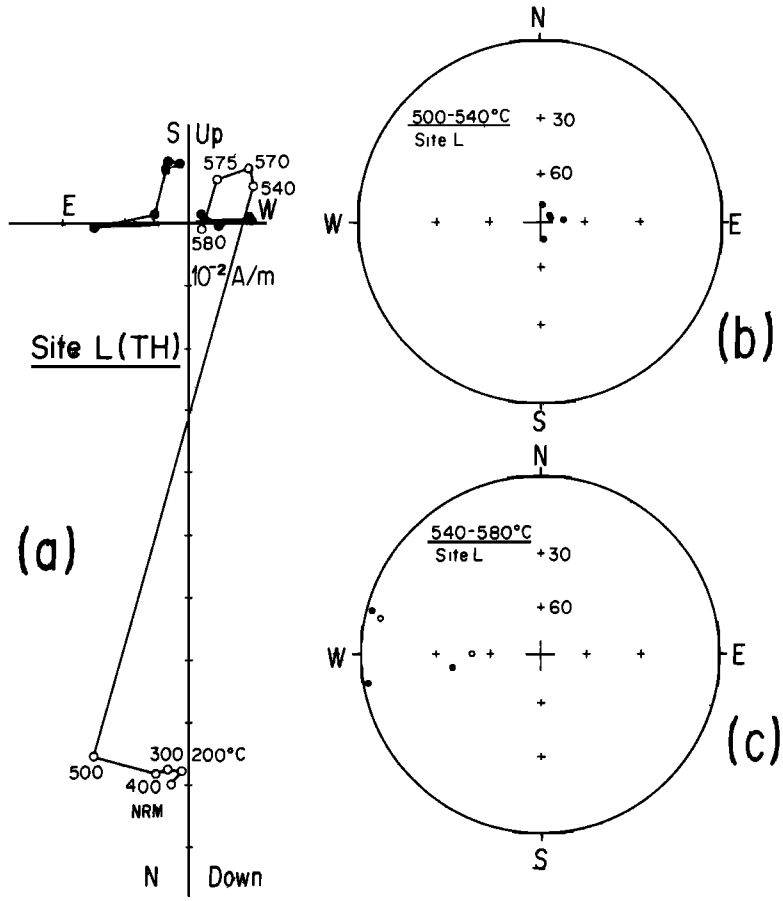


Fig. 9. Thermal demagnetization results from site L from the Saint-Varent area. (a) Zijderveld diagram of a thermal demagnetization of a sample containing both upward and downward characteristic directions, (b) stereographic plot of the downward directions obtained in the temperature range of 500°-540°C, and (c) stereographic plot of the imperfectly isolated upward or intermediate directions, obtained in the temperature range of 540°-580°C.

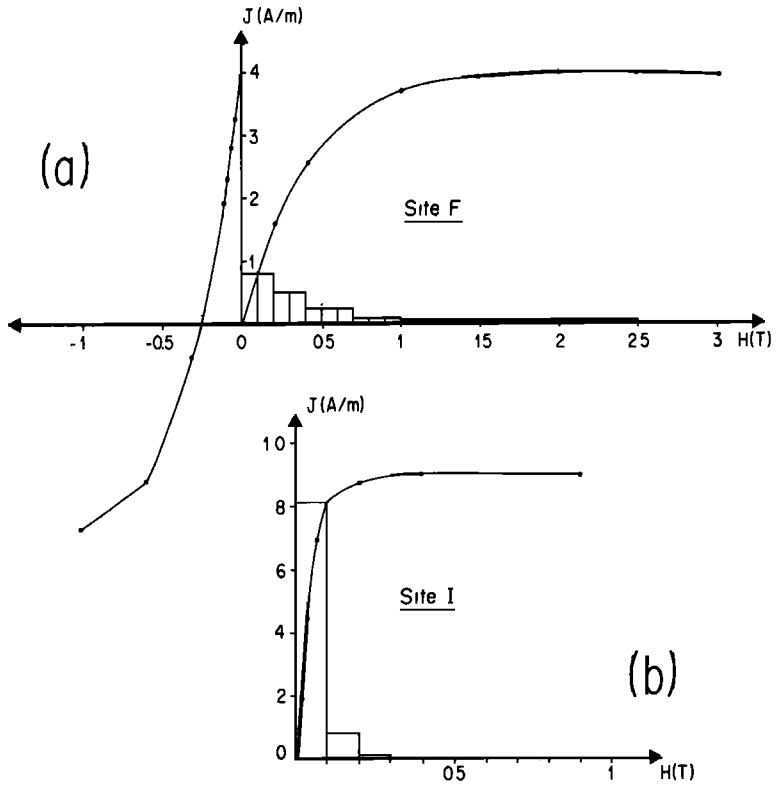


Fig. 10. Isothermal remanent magnetization (IRM) acquisition curves for (a) a sample from site F containing hematite and (b) a sample from site I containing only magnetite. In Figure 10a the backfield demagnetization of the IRM is also shown.

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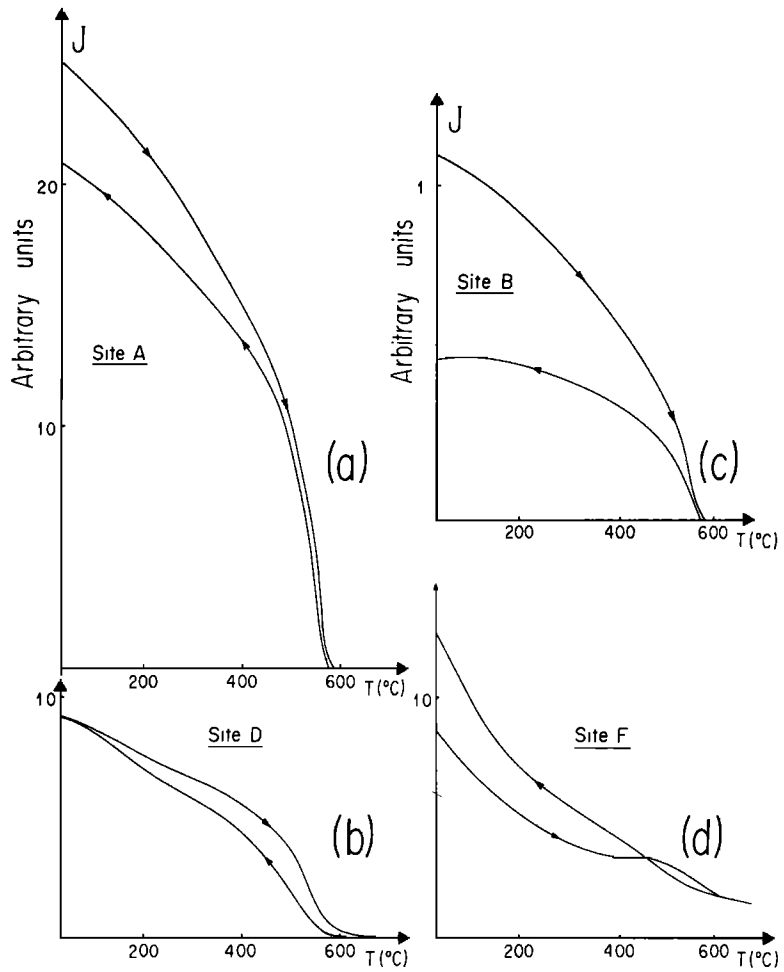


Fig. 11. Thermomagnetic curves indicative for (a,b,c) magnetite or (d) possibly hematite. In addition, the sample shown in Figure 11c reveals mineralogical changes recognized in the irreversibility of the curves.

Ordovician inclinations in Armorica, discussed in the next section, we do not believe that any indications for structural tilt exist.

Thus we conclude that the direction of 340°/-83° accurately represents the Late Ordovician paleofield, and that the paleopole at 34°N, 5°E is reliable.

The secondary magnetizations, determined through vector subtraction on three or more linear points and compiled in Table 2, yield a mean direction of 219°/+20° and a paleopole at 23°S, 316°E. The prominence of this magnetization near the Carboniferous shear zone, and the agreement between this pole and the average Carboniferous pole for the Armorican Massif (30°S, 326°E; Perroud et al. [1985]), argues for a Carboniferous age, associated with the Hercynian orogenic phases.

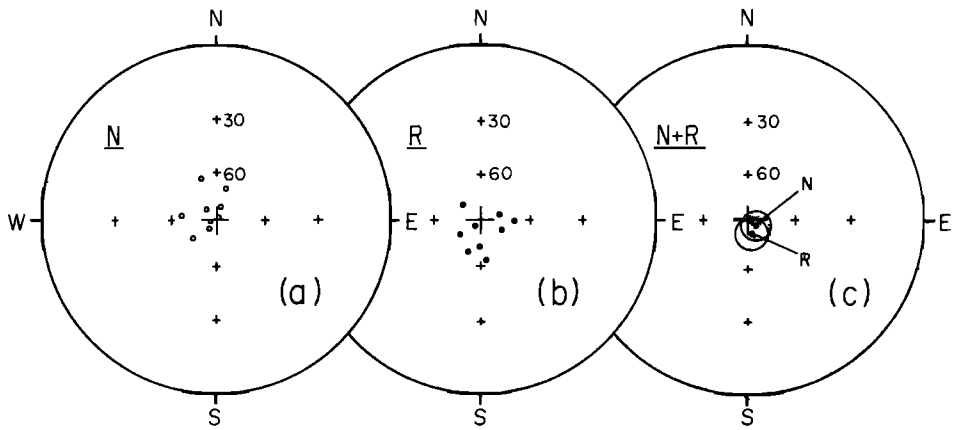


Fig. 12. Stereographic plots with the site-mean directions for (a) upward directions, (b) downward directions, and (c) their respective means with their cones of 95% confidence. In the last plot, the upward mean direction has been inverted to the corresponding downward one for easy comparison.

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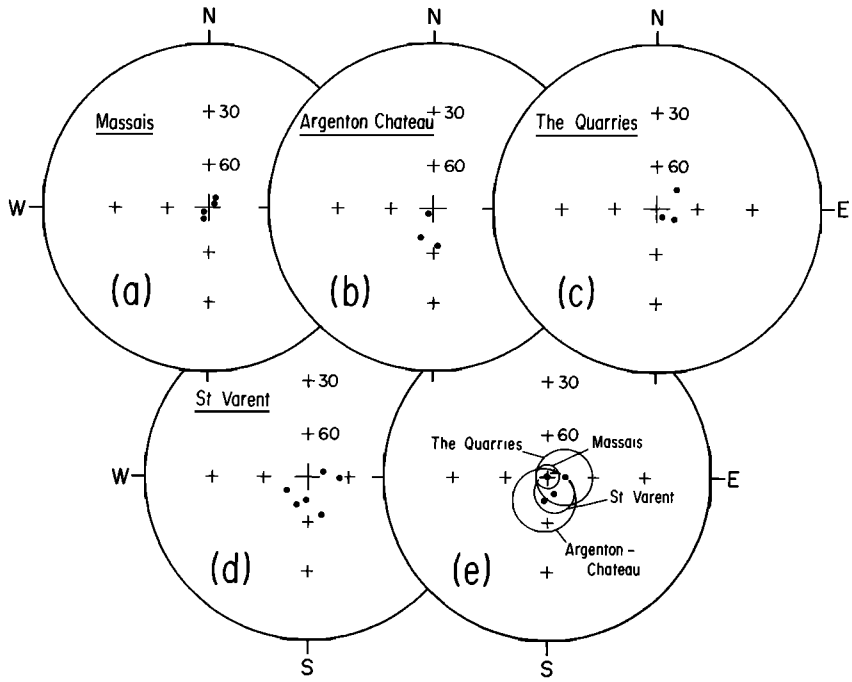


Fig. 13. Stereographic plots of site-mean directions (gabbroic and granodioritic sites only) for (a) the Massais area (sites A, C, D, E), (b) the Argenton Chateau area (sites F, G, H), (c) the quarries (sites I, J), and (d) the Saint-Varent area (sites K, L, M, N, O), as well as (e) the means of these four groups in the same plot with their cones of 95% confidence. Site mean directions are given in Table 1, and upward directions are inverted here to downward directions.

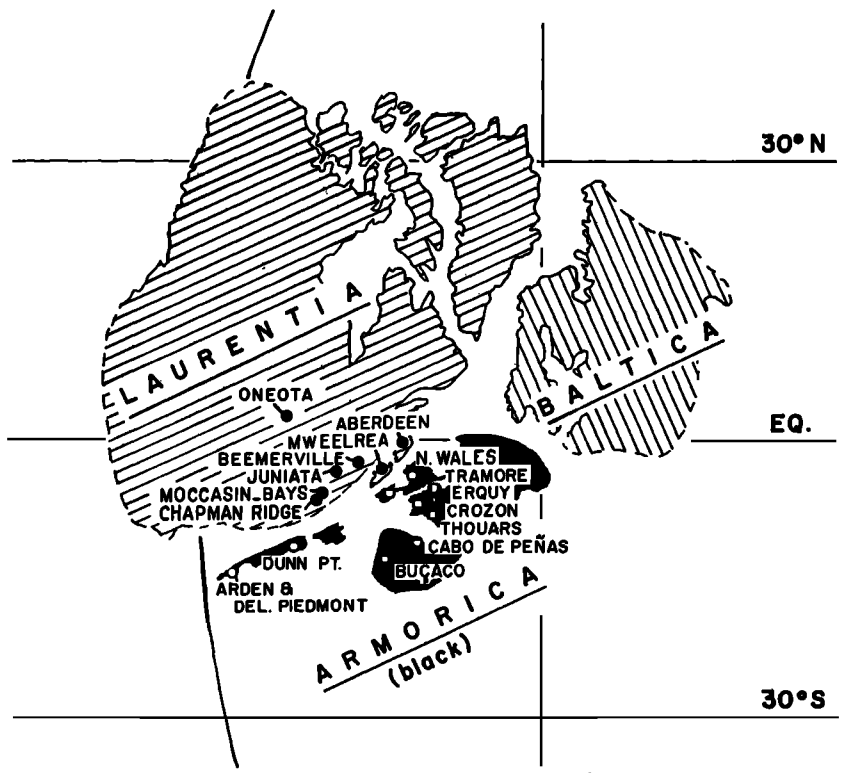


Fig. 14. Paleogeographic map of the Devonian Old Red continent, showing the tectonic elements Laurentia, Baltica, and Armorica in different signatures, with for Laurentia and Armorica the localities indicated where Ordovician paleomagnetic results have been obtained (Table 3). These Ordovician results indicate that at that time Laurentia and Armorica were at very different paleolatitudes and separated by an Early Paleozoic ocean. Baltica was presumably separate as well in the Ordovician. The three continents are thought to have collided in the Late Silurian to Early Devonian, causing the Caledonian and Acadian orogenies in between them.

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TABLE 3. Ordovician Paleopoles Discussed in the Text

Formation	Age	Pole Coordinates	Paleo-latitude	Reference
The North American Craton				
Beemerville	Ou (440 Ma)	35S, 306E	11.5S	Proko and Hargraves [1973]
Juniata	Ou	32S, 294E	17.1S	Van der Voo and French [1977]
Moccasin/Bays	Om	33S, 327E	6.5S	Watts and Van der Voo [1979]
Chapman Ridge	Om	27S, 292E	25.7S	Watts and Van der Voo [1979]
Oneota Dolomite	Ol	11S, 345E	1.7S	Jackson et al. [1983]
Other Possibly- North American Areas				
N. Ireland-Mweelrea	Om	11S, 38E	13.8S	Deutsch [1980, 1984]
Scotland-Aberdeen 1	Ou (441 Ma)	5S, 329E	23.4S	Watts and Briden [1984]
-Aberdeen 2	Ou (446 Ma)	0S, 7E	32.0S	Watts and Briden [1984]
-Aberdeen 3	Ou (448 Ma)	11S, 353E	21.4S	Watts and Briden [1984]
-Aberdeen 4	Om (468 Ma)	14S, 32E	13.0S	Watts and Briden [1984]
-Aberdeen 5	Om (>468 Ma)	18S, 32E	9.2S	Watts and Briden [1984]
The Ibero-Armorican Arc (Armorica)				
Thouars Massif	Ou (444 Ma)	34N, 5E	76.4S	This study
Erquy Spilites	Ou	35N, 34E	60.1S	Duff [1979]
Crozon	Ou	3N, 358E	45.6S	Perroud et al. [1983]
Buçaco	Ou	25N, 335E	69.6S	Perroud and Bonhommet [1981]
Cabo de Peñas	Ou	30N, 330E	66.5S	Perroud and Bonhommet [1981]
Other Possibly- Armorican Areas				
Delaware Piedmont	Ou (440 Ma)	48N, 288E	81.2S	Brown and Van der Voo [1983]
Arden, Delaware	Ou (440 Ma)	16N, 303E	61.1S	Rao and Van der Voo [1980]
Nova Scotia-Dunn Pt.	Ou	10N, 300E	53.5S	Johnson and Van der Voo [1983]
S. Ireland-Tramore	Om	11N, 18E	44.0S	Deutsch [1980, 1984]
N. Wales-granites	O	68N, 288E	55.6S	Thomas and Briden [1976]
N. Wales-basalts	O	73N, 49E	60.4S	Thomas and Briden [1976]

O, Ordovician; u, upper; m, middle; and l, lower. The paleolatitudes are computed directly from the inclinations for the site location of the study.

Geodynamic Implications

Previously published Ordovician results from Europe and areas adjacent to it in a late Paleozoic reconstruction of the northern continents [Van der Voo, 1983] reveal that for the North American and Baltic cratons, low paleolatitudes (i.e., low inclination values) prevail. In contrast, an area predominantly in Hercynian Europe can be outlined with high (>50°) paleolatitude values (Figure 14 and Table 3).

It must be noted, however, that none of the previous high-paleolatitude studies in Hercynian Europe would pass stringent reliability criteria insofar as ages or magnetic directions are concerned. Of the results compiled in Table 3, the Erquy and Crozon poles have somewhat uncertain ages of magnetization, and the Buçaco and Cabo de Peñas results were part of complex multicomponent magnetizations without clear separation of components. For Wales, the authors themselves [Thomas and Briden, 1976] considered the results as anomalous. However, all these results, as well as the Tramore pole from southern Ireland [Deutsch, 1980, 1984] and the poles from the Avalon and Piedmont terranes in easternmost North America, can be included in a common outline of high Ordovician paleolatitudes (Figure 14); this area includes virtually all areas originally included by Van der Voo [1979] in his Armorica plate concept. Figure 14 contrasts, in a Late Paleozoic assembly of the Old Red continent [Van der Voo, 1983], the Early Paleozoic equatorial continents of North America (Laurentia) and Baltica with the near-polar continental element of Armorica.

Well-known Ordovician glacial relicts in the western Sahara [Dillon and Sougy, 1974], and in

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Hercynian Europe as well [Hamoumi et al., 1981; Dangeard and Doré, 1971; Doré and LeGall, 1972], argue for a juxtaposition of Africa and Armorica in the vicinity of the south pole. Ordovician paleomagnetic results for Gondwana [Morel and Irving, 1978] and faunal considerations [Whittington and Hughes, 1972; Turner and Tarling, 1982; Cocks and Fortey, 1982] further support this paleogeography.

We therefore conclude that, in the Ordovician, Gondwana and Armorica were joined near the south pole, Laurentia and Baltica were located near the equator, and oceans such as Iapetus and an Ordovician Medio-European ocean separated these continents. By middle Devonian time, however, Laurentia, Baltica, and Armorica had coalesced into the Old Red continent. This scenario rules out a Taconic collision between Laurentia and Armorica as previously postulated by Van der Voo [1979] because there is not enough time to have Armorica move from polar to near-equatorial paleolatitudes to collide in the Late Ordovician. Instead, the Caledonian and Acadian orogenies must be regarded as being the products of the Laurentia-Baltica, Laurentia-Armorica, and Armorica-Baltica collisions, if these continents are to be assembled by Late Devonian time (as mandated by the paleomagnetic data for that time). This geodynamic picture implies a rapid northward drift of Armorica with velocities of the order of 10 cm/yr between 440 and 370 Ma.

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